

ERGA Assembly Report

v24.10.15

Tags: ERGA-BGE

TxID	669229
ToLID	xoWirArge9
Species	Wirenia argentea
Class	Solenogastres
Order	Pholidoskepia

Genome Traits	Expected	Observed
Haploid size (bp)	1,295,744,130	988,417,968
Haploid Number	9 (source: ancestor)	19
Ploidy	2 (source: ancestor)	2
Sample Sex	unknown	unknown

EBP metrics summary and curation notes

Obtained EBP quality metric for pri: 5.7.Q61

The following metrics were automatically flagged as below EBP recommended standards or different from expected:

- . Observed Haploid size (bp) has >20% difference with Expected
- . Observed Haploid Number is different from Expected
- . Kmer completeness value is less than 90 for pri
- . BUSCO single copy value is less than 90% for pri
- . Assembly length loss > 3% for pri
- . More than 1000 gaps/Gbp for pri

Curator notes

. Interventions/Gb: 9999999
. Contamination notes: "No contaminations were detected with FCS-GX."
. Other observations: "This was a special project. There were 2 PacBio data set - both ultra low input. Several attempts to combine the data sets failed. In the end we only used PacBio data from individual xoWirArge9 for the hifiasm contig assembly. The primary collapsed contigs had following stats: 58004 contigs, 3.04Gb yield; n50 189K and a BUSCO of 86.59% with 61.55% duplication rate. Automatic purging with purge_dups was suboptimal. We lost Busco genes but duplication rate was still above 30%. Therefore we did an iterative purging approach and removed junk-, haplotig-, repeat-tagged contigs that don't have any Busco genes. This approach converged after eight rounds at: 3722 contigs, 1.2Gb yield, 519Kb n50 and a BUSCO of 85.25% with 16.47% duplication rate. The HiC data is based on several individuals and of low quality: 78% duplication rate and very few long range interaction: 437341 read pairs at cis_40kb+. Initial Yahs scaffolding did a very bad job - contig error correction

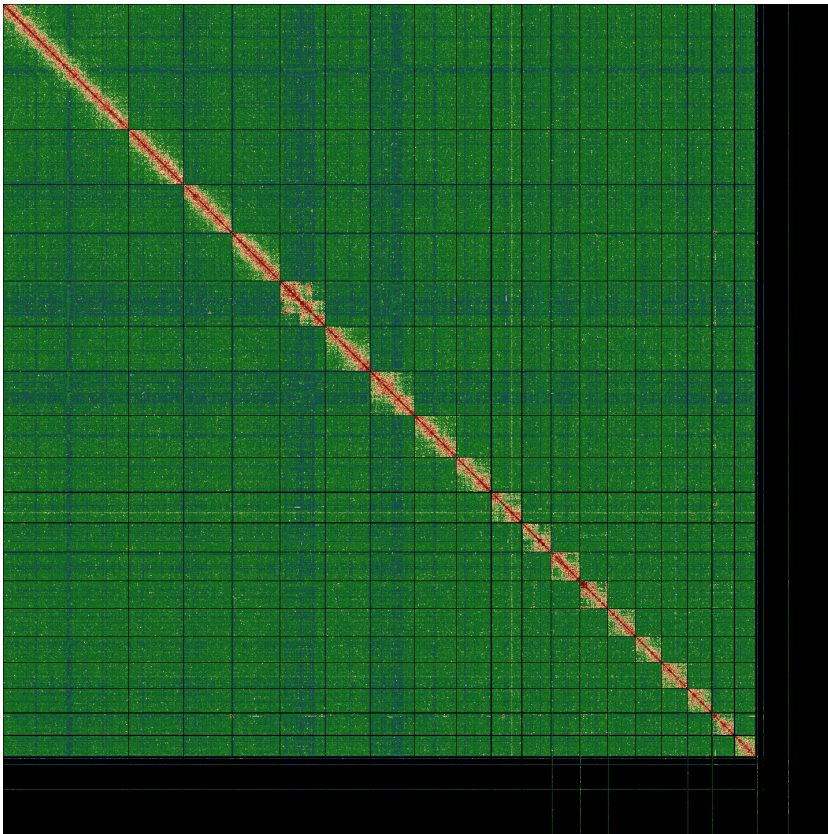
was disabled. Therefore we started the manual curation from the contigs but in the order and orientation proposed from yahs. In the first curation round all contigs were roughly assigned to putative chromosomes without splitting any contigs (because the resolution was simply too bad). The second curation round is based on smaller chunks, i.e. maps based on 2-3 chromosomes. This allowed for better detection of duplicated signals. In this step the contigs were trimmed according to the duplication HiC-signal. In order to decide which contigs should be marked as haplotig the Busco genes were checked and we tried to minimize the BUSCO duplication rate. So far no scaffolding was done. Afterwards all curated contigs which are sorted and oriented into putative chromosomes were combined and two more curation rounds were done. This time on the full assembly and we created scaffolds. This assembly is far from being great. But it would be nice to get a another opinion on it :) "

Quality metrics table

Metrics	Pre-curation pri	Curated pri
Total bp	1,203,184,737	988,417,968
GC %	39.85	39.87
Gaps/Gbp	0	2,300.65
Total gap bp	0	454,800
Scaffolds	3,722	925
Scaffold N50	519,909	49,430,100
Scaffold L50	748	8
Scaffold L90	2,232	19
Contigs	3,722	3,199
Contig N50	519,909	485,898
Contig L50	748	656
Contig L90	2,232	1,966
QV	60.8941	61.2222
Kmer compl.	71.1514	62.1923
BUSCO sing.	64.3%	82.8%
BUSCO dupl.	21.2%	2.2%
BUSCO frag.	3.1%	3.1%
BUSCO miss.	11.4%	11.9%

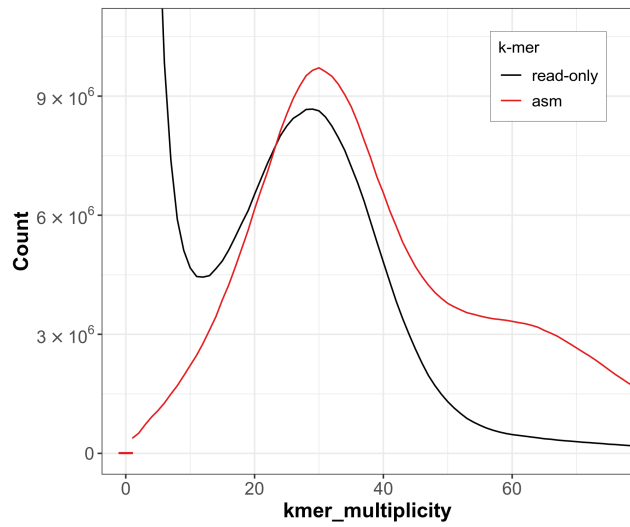
BUSCO: 6.0.0 (euk_genome_min, miniprot) / Lineage: mollusca_odb12 (genomes:36, BUSCOs:4421)

HiC contact map of curated assembly

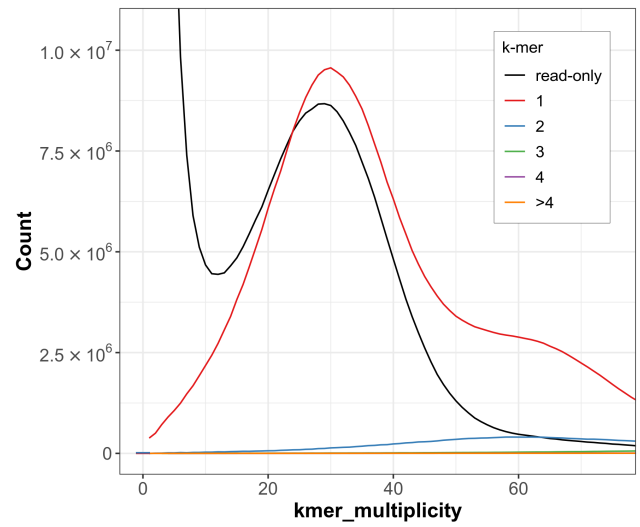


pr1 [\[LINK\]](#)

K-mer spectra of curated assembly

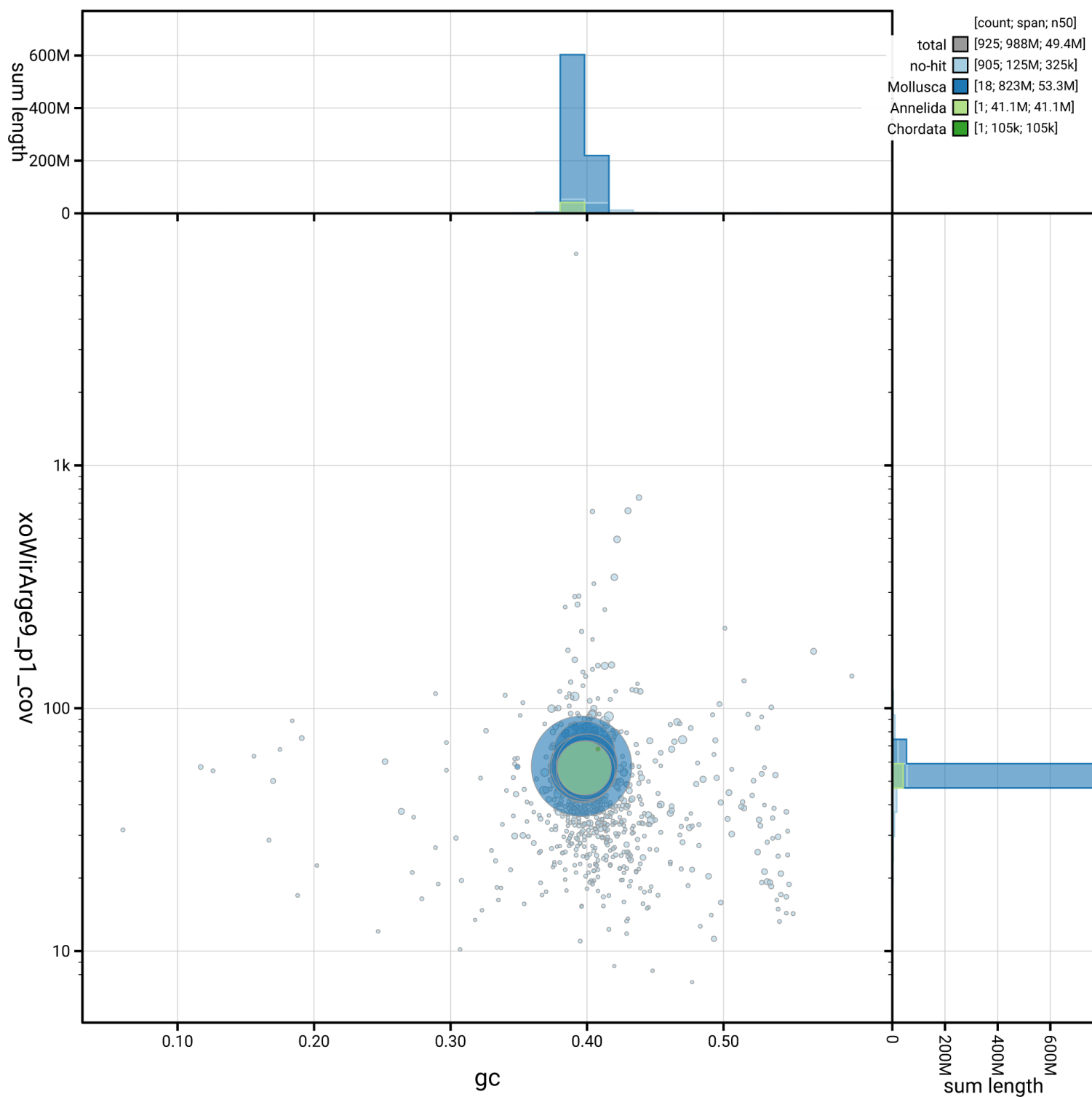


Distribution of k-mer counts coloured by their presence in reads/assemblies



Distribution of k-mer counts per copy numbers found in asm

Post-curation contamination screening



pri. Bubble plot circles are scaled by sequence length, positioned by coverage and GC proportion, and coloured by taxonomy. Histograms show total assembly length distribution on each axis.

Data profile

Data	HiFi	HiC
Coverage	29x	66x

Assembly pipeline

- **Hifiasm**
 - |_ *ver*: 0.25.0-r726
 - |_ *key param*: l2
- **purge_dups**
 - |_ *ver*: 1.2.6
 - |_ *key param*: NA

Curation pipeline

- **GRIT_Rapid**
 - |_ *ver*: 1a3d79a8
 - |_ *key param*: NA

Submitter: Martin Pippel

Affiliation: SciLifeLab

Date and time: 2026-04-29 09:57:38 CEST